

Every Trade Provable. Every Loss Capped.

Alpaca AI Trading Agents Hackathon — one-page write-up

Paper account PA38HG4D9653 · single-leg long 0DTE on SPY / QQQ / IWM · paper only, no live path in the repo.

Result: a fresh \$100,000 account finished at \$179,087 (+79.1%) across 3 trading sessions — 34 entries, 43 exits.

An autonomous agent that scans, scores, gates, sizes, executes and manages its own book with no human in the loop. Every control in it was validated against a year of real Alpaca OPRA option bars — 264 sessions, 4,544 alerts — and what did not survive that testing is not in the code. Repo: github.com/KIBA0993/Alpaca-Hackathon

1 · AI logic

A pure, unit-tested scorer reduces Alpaca 5-minute bars to one number from VWAP position, opening-range break, RSI, EMA stack and relative volume, plus a proposed direction. Two regime reads sit above it: a **half-opening-range noise band** (is this break larger than the symbol's own morning noise?) and **leader breadth** — 5 of 8 mega-caps above their 20-day average as of the last *completed* session, which only ever removes the opposed side.

Above that sits a **Claude reasoning layer scoped as a veto**. It argues bull and bear over the scored facts and returns strict JSON, and it can only turn a *go* into a *no-go* — it can never resurrect a trade the rules rejected. The deterministic path is therefore a strict subset of the AI path, and the two are auditable against each other. A missing key or an unparseable reply degrades to the deterministic gate and says so in the journal. The scored sessions ran the deterministic gate; the veto ships tested and runnable via `--decision-mode llm`.

2 · Risk gates

Single-leg long options only: maximum loss is the premium paid, enforced by the instrument rather than by code that has to remember. On top of that, `src/risk.py` and `src/execution.py` enforce, in order:

one-lot guard	never a second open lot on a (symbol, direction) already held
one direction / underlying	no simultaneous call <i>and</i> put on the same symbol
entry dedup	30 min, entry-anchored — a persistent score cannot stack the same name
adaptive sizing	target trimmed to the account's real options buying power, so an expensive contract sizes <i>down</i> instead of rejecting
session window	no entry after 15:00 ET; hard flatten 15:50 — a 0DTE never reaches expiry
orphan sweep	any option position the agent did not open is flattened at startup and EOD

Exits ladder in both directions: **+40% sells half** and trails the runner by 40% of its peak gain; **-20% sells half, -40% the rest**; a 30-minute stop closes anything still under water; 15:50 flattens the book. **174 network-free tests** cover the scorer, the gates, the sizing maths and the execution path.

3 · Alpaca infrastructure

Every order is placed through the Alpaca CLI, not the SDK (`src/broker_cli.py`): `alpaca order submit`, then poll to a terminal state — so the agent books the *real broker fill* rather than a quote, and cash P&L is buy-fill against sell-fill. Two safety belts: before placing anything it verifies that `alpaca doctor resolves` `paper-api.alpaca.markets` and refuses otherwise (the environment variable alone is not proof — a profile's `live_trade` can route live), and on an ambiguous submit failure it reconciles by `--client-order-id` via `alpaca order get-by-client-id` before it will ever resubmit, so a stalled order is never duplicated or orphaned. A paper order becomes a position only when it actually **fills**, and entries use a marketable limit to bound slippage.

Market data — bars, contract discovery, ATM quotes — comes from the Alpaca Market Data API. The agent runs unattended as an **isolated Docker container** under a supervisor that starts one session each weekday. Keys are injected at runtime, never baked into the image.

Every scan appends one JSONL record — score, each signal, band state, gate verdict, risk verdict, order — to `logs/decisions-YYYY-MM-DD.jsonl`. Most agents show you the trades they took; this one shows the trades it refused, and exactly why.

Paper trading only. Hypothetical results; not investment advice.